UK HealthCare
KENTUCKY CHILDREN'S HOSPITAL

Guideline/Protocol Title	Interim Pharmacotherapy Guidance for the Inpatient Management of SARS-CoV-2 (COVID-19) Infection in Pediatric Patients
Authors	Elizabeth Autry, PharmD, BCPPS Katie Olney, PharmD Laura Stadler, MSEd, MD, MS, FAAP Sean McTigue, MD Joel Howard, MD James Grubbs, MD Katie Wallace, PharmD Sarah Cotner, PharmD
Committee Review	Pharmacy and Therapeutics Committee
Target Population	Pediatric patients with confirmed COVID-19
Overview	This guideline discusses the warnings related to FDA approved as well as off-label and/or experimental use of medications based off of limited clinical data evidence for the management of COVID-19 in pediatric patients.
Effective Date	4/2020
Revised Date	**4/2023**
Expiration Date	10/2024
Schedule for Periodic Review	**6 months**
Implementation Strategy	Guideline will be shared on UK HealthCare COVID-19 Website
Education Strategy	N/A
Primary Outcome (s)	N/A
Outcome Assessment Plan	N/A
Information Technology Needs	N/A

Interim Pharmacotherapy Guidance for the Inpatient Management of SARS-CoV-2 (COVID-19) Infection in Pediatric Patients

Updated 04/20/2023

On April 25, 2022 the FDA approved the use of remdesivir in pediatric patients aged ≥28 days and weighing at least 3 kg who are positive for SARS-CoV-2 (COVID-19). As a result of remdesivir's FDA approval in pediatric patients aged ≥28 days, the agency also revoked the emergency use authorization for Veklury that previously covered this pediatric population. On June 15, 2020, the FDA revoked the EUA of hydroxychloroquine and chloroquine for the treatments of SARS-CoV-2 in light of ongoing serious cardiac adverse events and other series side effects. As literature surrounding the management of the novel coronavirus emerge, this document will be updated accordingly following the careful analysis of clinical trial study details.

Therapeutic Options for Pediatric Patients with SARS-CoV-2 (COVID-19)

This document provides guidance on the pharmacotherapy for SARS-CoV-2 confirmed infection, should it be deemed indicated by the treatment team. The current recommendations below are based on review of FDA approved treatment option(s), consensus guideline recommendations and published clinical data. As of April 25th, 2022, remdesivir is FDA approved for pediatric patients 28 days of age and older weighing at least 3 kilograms. This approval is based on phase II/III trial data evaluating remdesivir in pediatric patients for up to 10 days. **If there are questions or clarifications regarding a patient's disease classification and/or treatment needs, reach out to the Pediatric Infectious Diseases Provider On-Call.**

Remdesivir

Indication:

- Hospitalized patients with **CONFIRMED** (electronic or paper receipt of positive result from UKHC **OR** outside hospital) SARS-CoV-2 with severe or critical disease severity (or a hypothesized risk factor for severe SARS-CoV-2) (see **Appendix A**) admitted **within 7 days** of disease onset.
- Pediatric Infectious Disease Consult recommended for remdesivir therapy.

Dosing:

Remdesivir Loading Dose[a,b]	**Remdesivir Maintenance Dose**[a,b] (max duration of 5 days, regardless of disease severity)
5 mg/kg (maximum 200 mg) IV on day 1	2.5 mg/kg (maximum 100 mg) IV on days 2 through 5[a,b] **or until hospital discharge**, whichever is first

[a] If there are questions or concerns regarding order entry, please reach out to Katie Olney, PharmD (via EPIC Secure Chat), your primary team pharmacist OR the pediatric pharmacy satellite (3-6369).[b]

Monitoring Requirements:

- Baseline:
  - CMP for renal and hepatic function monitoring
    - HEPATIC Considerations: Assess for risk verse benefit if prior to therapy administration, ALT is ≥ 5 times the ULN or in patients with ALT elevations associated with elevated conjugated bilirubin, alkaline phosphatase, or INR
    - RENAL Considerations: Assess for risk verse benefit if prior to therapy administration, the patient has evidence of renal dysfunction, defined as:
      - Patients aged > 28 days with an eGFR < 30 mL/min or
      - Term neonates (7 to 28 days of life) with a serum creatinine ≥ 1 mg/dL
  - Prothrombin Time
- Daily:
  - CMP for renal and hepatic function monitoring
    - Consider discontinuation of therapy if ALT is ≥ 10 times ULN or ALT elevation is accompanied by signs or symptoms of liver inflammation
  - Prothrombin time

Other Resources:

- Drug Information: [Remdesivir [Veklury] Package Insert]

Dexamethasone

Indication:

- Hospitalized patients with **CONFIRMED** SARS-CoV-2 with severe or critical disease severity (see **Appendix A**)

Dosing:

- Dexamethasone 0.15 mg/kg (maximum 6 mg) IV or PO once daily
    1. Duration: up to 10 days or until hospital discharge, whichever comes first
    2. A corticosteroid taper may be considered based on patient response and duration of therapy

Alternatives Corticosteroids: to be considered if dexamethasone is not available

- Prednisone/Prednisolone 1 mg/kg (maximum 40 mg) PO once daily
- Methylprednisolone 0.8 mg/kg (maximum 32 mg) IV/PO once daily
- Hydrocortisone 3.75 mg/kg (maximum 160 mg) IV/PO once daily

Baricitinib

Indication:

- Hospitalized patients ≥2 years of age with **CONFIRMED** SARS-CoV-2 who meets all of the following inclusion criteria:
    1. Hospital admission ≤ 10 days
    2. ≥ 1 elevated inflammatory marker (Ferritin >400 ng/mL, D-dimer >0.5 ng/mL FEU, LDH >250 U/L, or CRP >8 mg/L)
    3. Rapidly increasing oxygen needs in patients requiring oxygen delivery through a high-flow nasal cannula (HFNC) or noninvasive ventilation (NIV)
- Pediatric Infectious Disease Consult recommended for baricitinib therapy.

Exclusion:

- Hospitalization > 10 days
- Patients on mechanical ventilation (MV) or extracorporeal membrane oxygenation (ECMO)
- Patients who have end-stage renal disease (ESRD), acute kidney injury, or who are on dialysis
- Patients with eGFR <15 mL/min/1.73 m^2
- Patients with severe hepatic impairment if potential risks outweigh potential benefits
- Patients on other JAK inhibitors or biologic disease modifying anti-rheumatic drugs (DMARDs)
- Patients receiving steroids at a dose greater than the equivalent of dexamethasone 6 mg
- Patients with known active tuberculosis
- Patient with active serious infections other than COVID-19 or chronic/recurrent infections where the potential risks outweigh potential benefits

Ordering:

- See **Appendix C** for ordering process if inclusion and exclusion criteria met.
- Acknowledgement of a **Legal Warning Alert** is required in Epic upon order entry.
- Serious adverse events and medication errors must be reported. See **Appendix B** for the institutional COVID-19 Emergency Use Authorization Adverse Event Reporting Process.

Baricitinib Dosing:

Age	Dose
2 to <9 years	2 mg every 24 hours for up to 14 days, **or until hospital discharge**, whichever is first[a]
≥9 years	4 mg every 24 hours for up to 14 days, **or until hospital discharge**, whichever is first[a]

[a] If there are questions or concerns regarding order entry, please reach out to Katie Olney, PharmD (via EPIC Secure Chat), your primary team pharmacist OR the pediatric pharmacy satellite (3-6369).

Monitoring Requirements:

- Baseline:
  - QuantiFERON-TB Gold
    - Result is **NOT** required prior to initiation of baricitinib.
    - Patient history should be obtained, chart review for previous TB screenings should occur, and a QuantiFERON-TB Gold result may be ordered.
  - CMP
  - CBC with differential
- Daily:
  - CBC with differential
  - CMP

Baricitinib Dose Adjustments for Patients with Abnormal Laboratory Values:

Laboratory Parameter[a,b]	Value	Recommendation
eGFR	≥60 mL/min/1.73 m^2	2 to < 9 years: 2 mg Q24h
		≥9 years: 4 mg Q24h
	30 to <60 mL/min/1.73 m^2	2 to < 9 years: 1 mg Q24h
		≥9 years: 2 mg Q24h
	15 to <30 mL/min/1.73 m^2	2 to < 9 years: NOT RECOMMENDED
		≥9 years: 1 mg Q24h
	<15 mL/min/1.73 m^2	NOT RECOMMENDED
Absolute Lymphocyte Count (ALC)	≥200 cells/µL	Maintain dose
	<200 cells/µL	Consider interruption until ALC is ≥200 cells/µL
Absolute Neutrophil Count (ANC)	≥500 cells/µL	Maintain dose
	<500 cells/µL	Consider interruption until ANC is ≥500 cells/µL
Aminotransferases	If increases in ALT or AST are observed and drug-induced liver injury (DILI) is suspected	Interrupt baricitinib until the diagnosis of drug induced liver injury is excluded

[a] Abbreviations: ALC = absolute lymphocyte count, ALT = alanine transaminase, ANC = absolute neutrophil count, AST = aspartate transaminase, eGFR = estimated glomerular filtration rate. [b] If a laboratory abnormality is likely due to the underlying disease state, consider the risks and benefits of continuing baricitinib at the same or a reduced dose.

Other Resources:

- Drug Information: [Baricitinib [Olumiant] Package Insert]
- HealthCare Provider Fact Sheet: https://www.fda.gov/media/143823/download
- Patient or Caregiver Fact Sheet: https://www.fda.gov/media/143824/download

Other Therapies and Clinical Recommendations

Therapy	Pediatric Infectious Diseases Society Panel Recommendation
Antiviral	
Hydroxychloroquine (or Chloroquine)	• The panel recommends against the use of hydroxychloroquine, alone or in combination with azithromycin, for the treatment of COVID-19, outside of a clinical trial
Lopinavir-Ritonavir	• The panel recommends against the use of lopinavir-ritonavir, along or in combination with ribavirin, except as part of a clinical trial
Immunomodulatory	
General Statement:	*Suggest that immunomodulatory therapy only be used for pediatric patients in the setting of confirmed critical COVID-19* ***with*** *evidence of hyperinflammation. There are no immunomodulators with proven efficacy for the treatment of COVID-19 in pediatric patients, therefore no guidance can be provided to support the use of one immunomodulatory therapy over another*
Convalescent Plasma	• Use of convalescent plasma in pediatric COVID-19 may be considered as part of the established FDA emergency investigational new drug program
IVIG	• Do not recommend the use of IVIG for treatment of acute COVID-19 in pediatric patients
Tocilizumab (IL-6)	• IL-6 inhibition may be considered in the care of pediatric patients with critical COVID-19, with priority given to clinical trial enrollment if available
Anakinra (IL-1)	• IL-1 inhibition may be considered in the care of pediatric patients with critical COVID-19, with priority given to clinical trial enrollment if available
Interferons	• Type I or Type III interferon should not be used for pediatric COVID-19 patients outside of clinical trials

Anticoagulation in Pediatric Patients with Confirmed SARS-CoV-2:

<table>
<tr><th>Indication</th><th>Agent</th><th>Duration</th></tr>
<tr><td rowspan="2">Prophylaxis:
• Confirmed SARS-CoV-2 by PCR and
• One or more HA-VTE risk factors (see Appendix D)</td><td>Enoxaparin
• < 2 months: 0.75 mg/kg SC every 12 hours
• ≥ 2 months of age and BMI < 40:
  ○ ≤ 60 kg: 0.5 mg/kg SC every 12 hours
  ○ > 60 kg: 40 mg SC every 24 hours
• ≥ 2 months of age and BMI ≥ 40: 40 mg SC every 12 hours</td><td rowspan="2">Can discontinue at discharge</td></tr>
<tr><td>Heparin: in setting of renal dysfunction (estimated GFR < 30 mL/min/1.73 m^2)
• > 60 kg: 5000 mg SC every 8 to 12 hours
• BMI ≥ 40: 7500 mg SC every 8 to 12 hours</td></tr>
<tr><td colspan="3">For patients that do not meet the above criteria, anticoagulation therapy should be tailored to the patient's risk of thrombosis. Please contact Pediatric Hematology with any questions or patient specific needs.
See Appendix E for contraindications to anticoagulation therapy</td></tr>
</table>

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Vaccine Recommendations in Pediatric Patients 6 Months to 18 Years of Age:

- Two COVID-19 vaccines are available in the United States for pediatric patients <18 years of age: Pfizer-BioNTech and Moderna. Please refer to updated CDC guidance for COVID-19 vaccine recommendations in pediatrics: https://www.fda.gov/news-events/press-announcements/coronavirus

COVID-19 Vaccination and SARS-CoV-2 Infection:

- Prior COVID-19 Infection: vaccines can be given safely to people with prior SARS-CoV-2 infection
- Current COVID-19 Infection: defer vaccination until recovered from the acute illness and criteria has been met for discontinuing isolation
    - There is **no** recommended minimum interval between infection and vaccination.

Appendix A: SARS-CoV-2 Disease Severity

Disease Severity	Definition
Mild	No new or increased from baseline supplemental oxygen requirement with symptoms limited to upper respiratory tract
Moderate	No new or increased from baseline supplemental oxygen requirement with symptoms involving the lower respiratory tract OR radiographic finding on chest radiograph
Severe[a]	New or increased from baseline supplemental oxygen requirement without the need for new or increase in baseline noninvasive/invasive mechanical ventilation[b]
Critical	New or increased requirement for invasive or noninvasive mechanical ventilation[b], ECMO, sepsis or multiorgan failure OR rapidly worsening clinical trajectory that does not yet meet these criteria
[a] Consideration for severe disease could be made based on a sustained SpO_2 of $\leq$ 94% on room air [b] noninvasive mechanical ventilation includes high-flow nasal cannula, continuous positive airway pressure or bilevel positive airway pressure	

Underlying Condition or Characteristic	Consideration for Antiviral Therapy
Medical Complexity	*Presence of one of these conditions or characteristics could be considered when making antiviral treatment decisions in the setting of a risk of severe SARS-CoV-2 infection*
Older Age	
Severe Immunocompromise	
Severe Underlying Cardiac Disease	
Severe Underlying Pulmonary Disease	
Obesity	

Appendix B: COVID-19 Emergency Use Authorization Adverse Event Reporting Process

Applies to agents issued an emergency use authorization for prophylaxis or treatment of COVID-19. For fully FDA approved COVID-19 therapeutics [e.g., remdesivir for the treatment of COVID-19 in adults and pediatric patients (28 days of age and older and weighing at least 3 kg)], it is up to the provider to determine clinical necessity of reporting internally and to [MedWatch](); please follow the [Adverse Drug Reaction reporting policy #A14.150]().

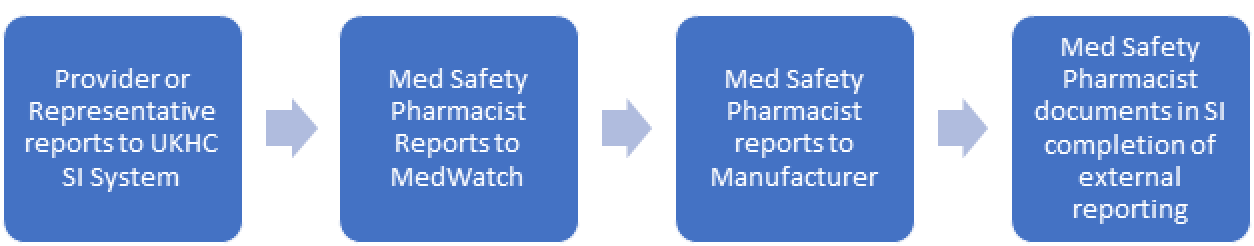
Mandatory reporting to FDA MedWatch <u>within 7 calendar days from the onset or awareness of the event</u> is required in 2 scenarios for COVID-19 drug administration under emergency use authorization (EUA).

Scenario	Responsible Party
1. Medication errors whether or not associated with an adverse event	Primary Provider or Representative
2. Serious adverse events (irrespective of attribution to vaccination), defined as: • Death • A life-threatening adverse event • Inpatient hospitalization or prolongation of existing hospitalization • A persistent or significant incapacity or substantial disruption of the ability to conduct normal life functions • A congenital anomaly/birth defect • An important medical event that based on appropriate medical judgement may jeopardize the individual and may require medical or surgical intervention to prevent one of the outcomes listed above	Primary Provider or Representative

Reporting Process

In the case that a medication error occurs or a recipient of an EUA drug experiences a side effect, adverse drug reaction, or adverse drug event, the following should occur:

1. Provider, administering nurse, or other team member identifies medication error or serious adverse event.
2. Provider or Representative reports to internal UKHC reporting system under the event type 'Adverse Drug Reaction' or 'Medication Event' as appropriate using template below. - http://careweb.mc.uky.edu/psn/
3. Once reported internally, the Med Safety Pharmacist (Liz Hess) submits MedWatch form to FDA - https://vaers.hhs.gov/reportevent.html. Reporter will be contacted if there are additional questions.
4. Med Safety Pharmacist attaches MedWatch Report to SI Report.
5. Med Safety Pharmacist reports adverse event to manufacturer online as needed and attaches to SI report.



Recommended COVID EUA Reporting Template
Use this template to report internally (copy/paste) or write down pertinent details, prior to internal reporting. Delete text that is not utilized (e.g. COVID hospitalization).
Patient Name: ___________________________
DOB: ___/_____/_______
Patient phone: __________________________
Patient email:___________________________
Allergies (drug/food and reaction):____

Date(s) of Administration: ________
Dose and Frequency: _____
Route of Administration: ____
Manufacturer: ______________
Lot #: ___________
Clinic or Unit Administering Drug: ___________
Injection site (if applicable):_________
Description of event/reaction: ____

Date of Clinic Visit or Hospitalization: ____/_______/2022
Reason for clinic visit or hospitalization: _______________
COVID-19 positive test result: Yes or No; if Yes, date ____/____/2022
Plans to monitor (include medications if prescribed):__________________________________

Appendix C: Baricitinib Ordering Process

Process	**Responsible Party**
Patient deemed candidate for baricitinib by primary team and infectious diseases attending, according to institutional criteria. Provider reviewed HealthCare Provider Fact Sheet for use of baricitinib under EUA. Baseline lab evaluation completed.	PRIMARY TEAM
Patient or patient's legal representative counseled on use of baricitinib and provided the Patients, Parents, and Caregivers Fact Sheet.	PRIMARY TEAM
Enter order. A legal alert warning/attestation acknowledgement is required upon order entry.	PRIMARY TEAM
Ensure patient meets criteria for use prior to order verification.	PRIMARY TEAM PHARMACIST OR PEDIATRIC SATELLITE PHARMACIST
Mandatory reporting required if patient experiences a serious adverse event potentially attributable to baricitinib, report via UK SI Reporting System.	PRIMARY TEAM

Appendix D: General Considerations

- Co-Infection:
    - Recommend to obtain a comprehensive viral respiratory panel to evaluate for possible co-infection
- Influenza Vaccine:
    - Prolonged administration of high doses of corticosteroids (i.e. a dose of prednisone of either 2mg/kg or greater or a total of 20 mg/day or greater or an equivalent) may impair antibody response
    - Influenza immunization can be deferred temporarily during the time of receipt of high dose corticosteroids, provided deferral does not compromise the likelihood of immunization before the start of influenza season

Appendix E: Risk Factors for Hospital-Associated VTE (HA-VTE) in Children

- Central venous catheter
- Mechanical ventilation
- Prolonged length of stay (i.e., anticipated > 3 days)
- Complete immobility (i.e., Braden Q Mobility Score = 1)
- Obesity (i.e., BMI > 95th percentile)
- Active malignancy, nephrotic syndrome, cystic fibrosis exacerbation, sickle cell disease vaso-occlusive crisis, or flare of underlying inflammatory disease (i.e., lupus, juvenile idiopathic arthritis, inflammatory bowel disease)
- Congenital or acquired cardiac disease with venous stasis or impaired venous return
- Previous history of VTE
- First-degree family history of VTE before age 40 years or unprovoked VTE
- Known thrombophilia (i.e., protein S, protein C, or antithrombin deficiency; factor V Leiden; factor II G20210A; persistent antiphospholipid antibodies)
- Pubertal, post-pubertal or age > 12 years
- Receiving estrogen-containing oral contraceptive
- Status-post splenectomy for underlying hemoglobinopathy

Appendix F: Contraindications to VTE Prophylaxis in Children

- Active or potential bleeding
- Coagulopathy/severe liver disease
- Heparin induced thrombocytopenia (HIT)
- Platelet count < 20,000 without coagulopathy
- Imminent/urgent surgical procedure
- Lumbar puncture, spinal injection or epidural catheter removal within past 2 hours
- Multiple trauma with high bleeding risk

References:

Consensus Guidelines:

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2. Dulek D, Fuhlbrigge R, Tribble A et al. Multidisciplinary Guidance Regarding the Use of Immunomodulatory Therapies for Acute Coronavirus Disease 2019 in Pediatric Patients. *Journal of the Pediatric Infectious Diseases Society.* 2020. DOI: 10.1093/jpids/piaa098
3. COVID-19 Treatment Guidelines Panel. Coronavirus Disease 2019 (COVID-19) Treatment Guidelines. National Institutes of Health. Available at https://www.covid19treatmentguidelines.nih.gov/. Accessed 8 Nov 2020.

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2. Beigel JH, Tomashek KM, Dodd LE, et al. Remdesivir for the Treatment of COVID-19 – Final Report. *NEJM*. DOI: 10.1056/NEJMoa2007764
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Dexamethasone:

1. The RECOVERY Collaborative Group. Dexamethasone in Hospitalized Patients with COVID-19 – Preliminary Report. *NEJM.* DOI: 10.1056/NEJMoa2021436

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2. COV-Barrier Study medRxiv: https://www.medrxiv.org/content/10.1101/2021.04.30.21255934v2
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2. Committee on Infectious Diseases. COVID-19 Vaccines in Children and Adolescents. *Pediatrics*. 2021; 148(2): e20210523336.

Other:

1. Sheahan TP, Sims AC, Leist SR, et al. Comparative therapeutic efficacy of remdesivir and combination lopinavir, ritonavir, and interferon beta against MERS-CoV. Nat Commun. 2020;11(1):222.
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